

Year 2 Assessed Problems

Semester 1

Assessed Problems 5

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 12th November

at 17:00

Quantum Mechanics 2 - Problem Sheet 2

A particle of mass m is incident on a potential barrier of width L and height V_0 . The potential is zero on both sides of the barrier. The particle has energy E , which is **greater** than the height of the barrier, $E > V_0$.

Write down expressions for the wavenumbers and wavefunctions in each of the three regions of space (across the barrier and on each side of it).

[3]

By applying the appropriate boundary conditions at both of the edges of the barrier, derive a set of equations to relate the coefficients of the wavefunctions in the different regions.

[3]

Hence show that, if the height of the barrier is

$$V_0 = E - \frac{\pi^2 \hbar^2}{2mL^2},$$

then there is no reflection from the barrier for this particle.

[4]